

[Request]How much water does AI use?

Posted by u/Commercial_Job_867 • 0 points • 6 comments

Bassically i am annoyed by people saying that AI is bad because it uses water, so how much water does AI data centres use per day and how much would it change if everyone used googl given that there is 2.5 billion prompts a day?

Comments

u/[deleted] • 1 points • 2025-12-25 02:25 UTC

[removed]

u/Elfich47 • 3 points • 2025-12-24 23:11 UTC

the issue is the cooling towers for server farms often use *drinking water* to refill their towers. and that means getting access to drinking water. and some of the places where these server farms are are very water poor (like the American south west). the land is cheap there because water is scarce. and server farms evaporate water as fast as it can get it. and that means people who want to Sri k water have problems.

u/autophage • 10 points • 2025-12-24 20:56 UTC

The issue is that people are bad at separating what they mean.

Training fundamental models takes an enormous amount of energy, of water, of compute resources - of all sorts of things. But fundamental model training isn't the thing that happens when someone sends a prompt to an AI service; parsing and responding to that takes a much smaller amount.

Some people try to amortize the amount of resources used to train a fundamental model across all requests to that model, but that gets kinda complicated for a bunch of reasons that there isn't space to go into here.

The other difficulty is that when someone talks about an AI query "taking" an amount of water, that can mean a couple of different things. Water being used as coolant is not consumed so that it disappears - it goes back into the evaporative cycle. It's like when you drink water - some of it comes back out as pee, some comes out as sweat, some stays in your cells (though even that will ultimately end up back in the water cycle). Now, that isn't to say that *no* water is "consumed" - if water is used in a way that it gets polluted and then not treated properly, that's no good; if water is being used in excess of what's available, that may drive droughts in areas that used to end up with that water.

But my point is that it's all pretty complicated, and coming to nice round figures with which to do math requires a *lot* of clarifications. AI companies will tend to answer things with the version of figures that looks best for them, AI haters will tend to cherry-pick the versions that look the worst.

u/RollsHardSixes • 7 points • 2025-12-24 23:29 UTC

Just to be clear if a data center is pulling water and evaporating it from an aquifer that will rejoin the water cycle but raining down over the ocean 500 miles away is still draining a critical resource from a community and that is what I mean specifically

u/geeharee • 1 points • 2025-12-28 18:10 UTC

Anyone who says 'It goes back into the water cycle' is being hilariously dishonest

u/tadrinth • 4 points • 2025-12-24 20:44 UTC

I believe these posts have some decent analysis, though perhaps not the exact question you are asking:

<https://andymasley.substack.com/p/the-ai-water-issue-is-fake>

<https://andymasley.substack.com/p/empire-of-ai-is-wildly-misleading>

u/AutoModerator • 1 points • 2025-12-24 20:37 UTC

###General Discussion Thread

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